SUPPLEMENT

Data Extraction

The following data were independently extracted from the included articles: name of the first author, year of publication, type of data (e.g., continuous and dichotomous data), main outcomes, whether a meta-analysis was performed, statistical software and methods used (statistical modeling for meta-analysis and methods for adjusting 95% confidence interval), risk of bias assessment tool, methods for evaluating non-reporting bias (i.e., publication bias), and whether the study protocols were registered. When a meta-analysis was performed, studies were further classified into univariate vs. bivariate meta-analyses. Univariate meta-analysis focuses on single-effect size (e.g., mean difference, proportion, relative risk, odds ratio, hazard ratio), which is most commonly implemented in meta-analyses of intervention studies; bivariate meta-analysis focuses on dual-effect sizes (sensitivity and specificity) and the correlation between them, which is used in meta-analysis of diagnostic test accuracy (DTA).

Reviewing the PRSIMA 2020 Checklists

Two reviewers, who were familiar with the PRISMA 2009 guidelines, reached a consensus for each item in the checklists. For example, item #13a ("describe the processes used to decide which studies were eligible for each synthesis") was considered to be satisfied when a study reported how many articles were included for each synthesis and provided citations for the included articles. This is because most outcomes or types of interventions in radiology studies are generally not complex, thus they do not require an additional process for deciding which study is eligible for which outcome synthesis. Regarding DTA studies, we also used the PRISMA 2020 checklists instead of the PRISMA-DTA checklists; this is supported by the following comment in the PRISMA guidelines: "For DTA study, we recommend authors report their review in accordance with the recommendations in PRISMA 2020 along with the guidance specific to the extension" [1].

Data Analysis

We evaluated the completeness of the reporting of main texts and abstracts using the PRISMA 2020 checklists and abstract checklists, respectively. The reviewing process was conducted by one author and double-checked by a second author in case of ambiguity in the assessment. For each item in the checklists, we calculated the proportion of studies that met the recommendation in the guidelines.

Managing Within-Study Covariance

To manage within-study covariance, a single-effect size that best represents a research question may be chosen, discarding the rest of the outcomes reported in an individual study [2]. From a statistical point of view, a multivariate meta-analysis may be the preferred method over multiple univariate meta-analysis [3]. This method is particularly useful as it can jointly synthesize multiple-effect sizes while avoiding inflation of type I error that may occur when a separate meta-analysis is performed for each outcome [4]. However, despite the benefit of providing more precise estimates, a multivariate meta-analysis may not change the direction of the conclusion derived from a univariate meta-analysis [5]. A previous study demonstrated that multivariate meta-analysis is suitable when the magnitude of the correlation between outcomes is large (> 0.5 or < -0.5) [4]. Therefore, we suggest that authors report a potential risk of correlation between outcomes when pooling multiple-effect sizes and consider using multivariate meta-analysis when the correlation is presumed to be significant.

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Supplementary Table 1. Characteristics of the Included Studies

Study	Study Type	Type of Meta-Analysis	Type of Data	Main Outcome	Number of Included Studies
Kang et al., 2014 [6]	Systematic review	NA	Dichotomous data	Reporting quality	308
Kim et al., 2015 [7]	Systematic review	NA	DTA data	Diagnostic accuracy	14
Wang et al., 2016 [8]	Systematic review and meta-analysis	Univariate	Time-to-event data Dichotomous data	Efficacy and safety of intervention	6
Zhu et al., 2016 [9]	Systematic review and meta-analysis	Univariate	Time-to-event data Dichotomous data	Efficacy of intervention	8
Choi et al., 2018 [10]	Systematic review and meta-analysis	Univariate	Time-to-event data Dichotomous data	Efficacy and safety of intervention	13
Liao et al., 2018 [11]	Systematic review and meta-analysis	Bivariate	DTA data	Diagnostic accuracy	27
Wang et al., 2018 [12]	Systematic review and meta-analysis	Univariate	Time-to-event data Dichotomous data	Efficacy and safety of intervention	8
Kim et al., 2019 [13]	Systematic review and meta-analysis	Univariate	Time-to-event data Dichotomous data	Efficacy and safety of intervention	17
Kim et al., 2019 [14]	Systematic review	NA	Dichotomous data	Study quality	516
Chung et al., 2019 [15]	Systematic review	NA	DTA data	Diagnostic accuracy	12
Kim et al., 2019 [16]	Systematic review and meta-analysis	Univariate	Dichotomous data	Efficacy of intervention	34
Kim et al., 2020 [17]	Systematic review and meta-analysis	Univariate	Continuous data	Agreement and reliability between imaging methods	20
Kang et al., 2020 [18]	Systematic review	NA	Continuous data	HRQoL score	45
Suh et al., 2020 [19]	Systematic review and meta-analysis	Univariate	Time-to-event data	Reliability between overall survival and imaging surrogate markers	23
Park et al., 2020 [20]	Systematic review and meta-analysis	Bivariate	DTA data	Diagnostic accuracy	55
Suh et al., 2020 [21]	Systematic review	NA	Descriptive data*	Imaging protocol	16
Cho et al., 2020 [22]	Systematic review and meta-analysis	Univariate	Dichotomous data	Imaging finding	17
Kim et al., 2020 [23]	Systematic review and meta-analysis	Bivariate	DTA data	Diagnostic accuracy	13
Kim et al., 2021 [24]	Systematic review and meta-analysis	Univariate	Dichotomous data	Efficacy and safety of intervention	11
Lim et al., 2021 [25]	Systematic review and meta-analysis	Univariate	Dichotomous data	Diagnostic yield	22
Kim et al., 2021 [26]	Systematic review and meta-analysis	Univariate	Dichotomous data	Imaging finding	17
Ko et al., 2021 [27]	Systematic review and meta-analysis	Bivariate	DTA data	Diagnostic accuracy	20
Kim et al., 2021 [28]	Systematic review and meta-analysis	Univariate	Continuous data	Agreement and reliability between imaging methods	16
Kim et al., 2021 [29]	Systematic review and meta-analysis	Univariate	Dichotomous data	Efficacy and safety of intervention	4

*Descriptive data regarding MR protocols for patient selection and outcome assessment in 16 randomized controlled trials are presented in the study. DTA = diagnostic test accuracy, HRQoL score = heathy-related quality of life score, NA = not applicable

Supplementary Table 2. Statistical Methods in the Included Studies

Study	Statistical Software	Statistical Methods			Risk of Bias Assessment	Non-Reporting Bias Assessment	Protocol Registration
		Fixed-Effects Model	Random-Effects Model	95% CI Adjustment			
Kang et al., 2014 [6]	Meta-analysis not performed	NA	NA	NA	NA	NA	Not performed
Kim et al., 2015 [7]	Meta-analysis not performed	NA	NA	NA	QUADAS	Not reported	Not performed
Wang et al., 2016 [8]	RevMan 5.2	Mantel-Haenszel method	Dersimonian-Laird	Not reported	RoB	Funnel plot	Not performed
Zhu et al., 2016 [9]	Stata	Not reported	Not reported	Not reported	RoB (RCT) Newcastle-Ottawa Scale (non-RCT)	Funnel plot Egger test	Not performed
Choi et al., 2018 [10]	'Metafor' package in R	Inverse variance model	Restricted maximum-likelihood estimation	Not reported	Newcastle-Ottawa Scale	Funnel plot* Egger test	Not performed
Liao et al., 2018 [11]	Stata	NA	Bivariate random-effects model HSROC	Not reported	QUADAS-2	Deeks' funnel plot* Deeks' asymmetry test	Not performed
Wang et al., 2018 [12]	RevMan 5.2	Mantel-Haenszel method	Mantel-Haenszel method	Not reported	Revised Jadad scale (RCT) Newcastle-Ottawa Scale (non-RCT)	Not reported	Not performed
Kim et al., 2019 [13]	'Metafor' package in R	NA	Restricted maximum-likelihood estimation	Not reported	US National Institutes of Health Quality Assessment of Case Series Studies tool	Funnel plot Egger test	Not performed
Kim et al., 2019 [14]	Meta-analysis not performed	NA	NA	NA	NA	NA	Not performed
Chung et al., 2019 [15]	Meta-analysis not performed	NA	NA	NA	QUADAS-2	Not reported	Not performed
Kim et al., 2019 [16]	'Meta' and 'Metafor' package in R	NA	Dersimonian-Laird	Clopper-Pearson interval	QUADAS-2	Funnel plot† Egger test	Not performed
Kim et al., 2020 [17]	'Meta' package in R	NA	Dersimonian-Laird	Not reported	QUADAS-2	Funnel plot* Egger test	Not performed
Kang et al., 2020 [18]	Meta-analysis not performed	NA	NA	NA	ROBINS-I	NA	Not performed
Suh et al., 2020 [19]	Not reported	Not reported which model was used	Not reported which model was used	Not reported	Not reported	Not reported	Not performed
Park et al., 2020 [20]	'Meta' package in R	NA	Bivariate random-effects model	Not reported	Not reported	Not reported	Not performed
Suh et al., 2020 [21]	Meta-analysis not performed	NA	NA	NA	NA	NA	Not performed
Cho et al., 2020 [22]	'Meta' package in R	NA	Dersimonian-Laird	Not reported	RoBANS	Not reported	Not performed
Kim et al., 2020 [23]	'Midas' module in Stata 'Mada; package in R	NA	Bivariate random-effects model HSROC	Not reported	QUADAS-2	Deeks' funnel plot Deeks' asymmetry test	Not performed
Kim et al., 2021 [24]	'Meta' and 'Metafor' package in R	NA	Dersimonian-Laird Binomial-normal model	Knapp and Hartung method	NA	Funnel plot* Egger test	Not performed
Lim et al., 2021 [25]	'Meta' package in R	NA	Dersimonian-Laird	Not reported	QUADAS-2	Funnel plot Egger test	Not performed
Kim et al., 2021 [26]	'Meta' and 'Metafor' package in R	NA	Dersimonian-Laird	NA	US National Institutes of Health Quality Assessment of Case Series Studies tool	Funnel plot Egger test	Not performed
Ko et al., 2021 [27]	Stata	NA	Bivariate random effects model HSROC	NA	QUADAS-2	Deeks' funnel plot Deeks' asymmetry test	Not performed
Kim et al., 2021 [28]	'Meta' package in R	NA	Dersimonian-Laird Bivariate random-effects model	Knapp and Hartung method	QUADAS-2	Funnel plot Egger test	Not performed
Kim et al., 2021 [29]	'Meta' package in R	NA	Not reported	NA	RoBANS	Not reported	Not performed

*Four studies did not thoroughly assess the possible causes of the asymmetry in the funnel plots, †One study adjusted the asymmetry in the funnel plot, using a trim-and-fill method, to calculate the publication bias-adjusted pooled estimates. CI = confidence interval, NA = not applicable, QUADAS = Quality Assessment of Diagnostic Accuracy Studies, RCT = randomized controlled trial, RoB = Risk of Bias in randomized trials, RoBANS = Risk of Bias Assessment tool for Non-randomized Studies, ROBINS-I = Risk of Bias in Non-randomized Studies of Interventions

Supplementary Table 3. Potential Risk of Correlation between Outcomes in the Studies Reporting Multiple Pooled Results

Study	Potential Source of Correlation between Outcomes
Zhu et al., 2016 [9]	Pooled results of oncologic outcomes (overall survival rate, objective response rate) are presented in HCC patients treated with TACE plus ^{131}I-labelled metuximab Result synthesis included overall survival rate at multiple time points (1 and 2 years), which were correlated
Choi et al., 2018 [10]	Pooled results of oncologic outcomes (overall survival rate, cancer-specific survival rate) are presented in RCC patients treated with microwave ablation Result synthesis included overall survival and cancer-specific survival rate at multiple time points (1, 2, 3, and 5 years), which were correlated
Wang et al., 2018 [12]	Pooled results of oncologic outcomes (overall survival rate, recurrence-free survival rate) are presented in HCC patients treated with combined RFA and TACE Result synthesis included overall survival and recurrence-free survival rate at multiple time points (1, 3, and 5 years), which were correlated
Kim et al., 2019 [13]	Pooled results of oncologic outcomes (overall survival rate and Kaplan-Meier rate of time to progression) are presented in HCC patients treated with radioembolization vs. sorafenib Result synthesis included overall survival and Kaplan-Meier rate of time to progression at multiple time points (3, 6 months, 1, 2, and 3 years), which were correlated
Kim et al., 2020 [17]	Pooled agreements between CT and MRI in the measurement of right ventricular volume and functional parameters are presented Functional parameters (end-diastolic volume, end-systolic volume, systolic volume, or ejection fraction) are likely to be correlated with one another
Suh et al., 2020 [19]	Pooled correlations between overall survival and image-based surrogate endpoints in glioblastoma are presented Image-based surrogate endpoints (progression-free survival, 6-months and 12 months progression-free survival) are likely to be correlated with one another
Kim et al., 2021 [28]	Pooled agreements between ECG-gated vs. non-ECG-gated CT in evaluating coronary artery calcium severities are presented In a subgroup analysis of slice thickness, data for each slice thickness were considered differently even though the data were from the same patients Therefore, the results for each slice thickness are likely to be correlated

ECG = electrocardiography, HCC = hepatocellular carcinoma, RCC = renal cell carcinoma, TACE = transarterial chemoembolization

Supplementary Table 4. Example of a "Summary of Finding" Table (Numbers in the Table Are Not from Real Data)

Summary of Findings:

Radiofrequency Ablation Compared to Open Surgery for Osteoid Osteoma

Patient or population: osteoid osteoma
Setting: Treatment
Intervention: Radiofrequency ablation
Comparison: open surgery

Outcome № of participants (studies)	Relative effect (95% CI)	Anticipated absolute effects (95% CI)			Certainty	What happens
				Difference		
Recurrence rate № of participants: 1970 (15 observational studies)	**RR 0.41** (0.23 to 0.56)	7.1%	**2.9%** (1.6 to 4)	**4.2% fewer** (5.5 fewer to 3.1 fewer)	⨁⨁⨁◯ MODERATE [a,b]	Radiofrequency ablation likely results in a large reduction in the recurrence rate.
Complication rate № of participants: 1600 (12 observational studies)	**RR 0.74** (0.67 to 0.95)	8.9%	**6.6%** (5.9 to 8.4)	**2.3% fewer** (2.9 fewer to 0.4 fewer)	⨁⨁◯◯ LOW [c,d]	Radiofrequency ablation may slightly reduce the complication rate.

GRADE Working Group grades of evidence
High certainty: We are very confident that the true effect lies close to that of the estimate of the effect
Moderate certainty: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different
Low certainty: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect
Very low certainty: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect

a. Follow-up time after treatment varies between the included studies.
b. Significant study heterogeneity (I^2 = 65%) was observed in the meta-analysis. Tumor size was associated with study heterogeneity.
c. Definition of moderate to severe complication differed between the included studies.
d. Asymmetry in the funnel plot was observed (p value = 0.03).

*Risk in the intervention group (and its 95% CI) is based on the assumed risk in the comparison group and the relative effect of the intervention (and its 95% CI).
CI = confidence interval, RR = risk ratio